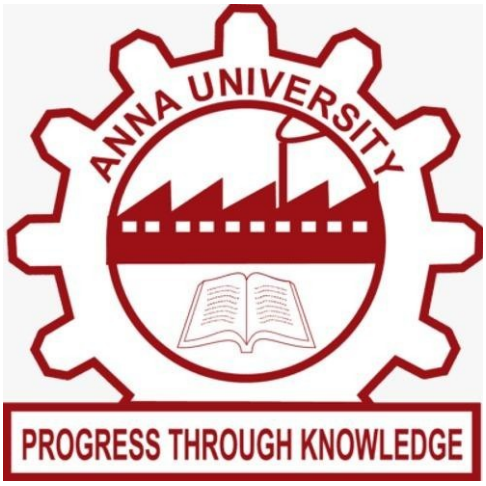
# UI/UX DESIGN OF ECOGUARD – POLLUTION CONTROL SYSTEM MOBILE APPLICATION

**A SUMMER INTERNSHIP REPORT**

*Submitted by*

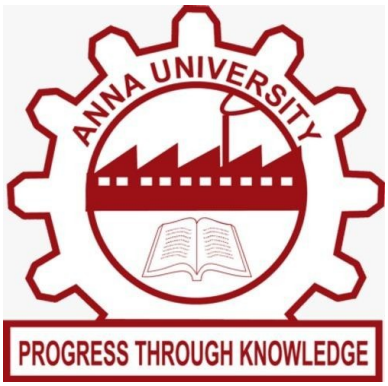
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# BONAFIDE CERTIFICATE

This is to certify that this Summer Internship project report "**UI/UX Design Of EcoGuard – Pollution Control System Mobile Application**" is the Bonafide work **of ARUVI V (810023104027)** and has satisfactorily completed the course **of CS3711 – SUMMER INTERNSHIP** for **SEVENTH semester, B.E. COMPUTER SCIENCE AND ENGINEERING** during the academic year **2026–2027**.

**Faculty Incharge** **Head of the Department**

Submitted for the practical examination held on _____.

# ABSTRACT

The EcoGuard – Pollution Control System is a UI/UX design project developed during my internship at Media Wave Technologies, Tiruchirappalli. The project focuses on designing a clear, calm, and trustworthy mobile experience under the tagline “Breathe. Monitor. Act.” It helps citizens check live air quality, understand pollutant levels, receive health alerts, and report pollution incidents, and it helps control-room operators monitor sensors and trigger control actions when pollution crosses safe limits.

The project follows a user-centered Design Thinking approach consisting of Empathize, Define, Ideate, Wireframing, Visual Design, Prototyping, and Testing. User research and competitor analysis were used to understand common issues in existing air-quality applications, such as data overload, unclear colour coding, and the absence of actionable advice. Based on the findings, user personas, a user flow, low-fidelity wireframes, a design system, high-fidelity screens, and an interactive prototype were created.

Figma was the primary design and prototyping tool. The final design uses a deep green and teal primary palette, the internationally recognised AQI colour scale for data, a soft white background, a clean sans-serif typeface, rounded cards and gauges, and a four-tab bottom navigation bar (Home, Map, Alerts, Profile). The application includes the Splash, Login, Sign Up, Home Dashboard, Pollutant Details, Monitoring Map, Alerts, Report Pollution, Control Panel, and Health Advisory screens.

Usability, responsive, functional, and visual checks were carried out on the prototype. The results indicated that the proposed interface improved clarity of air-quality information, reduced user effort in reporting incidents, and maintained visual consistency across screens. The internship provided practical exposure to professional UI/UX workflows and strengthened my skills in research, wireframing, design systems, prototyping, testing, documentation, and presentation.

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# 1.INTRODUCTION

An internship is an essential part of professional education because it connects academic learning with real-world industrial practice. During the internship at Media Wave Technologies, Tiruchirappalli, I worked in the UI/UX Design domain and designed a mobile application named EcoGuard for pollution monitoring and control. Air pollution is a growing public-health concern, yet the data published by many systems is technical and hard for ordinary people to act on. The project focused on presenting pollution data in a simple, visual, and actionable way. The design process was carried out primarily in Figma and followed a user-centered Design Thinking methodology.

The project included user research, competitor analysis, requirement analysis, user personas, user flow, low-fidelity wireframes, a design system, high-fidelity UI design, interactive prototyping, and usability testing.

## 1.1 OBJECTIVES OF THE INTERNSHIP

- To understand professional UI/UX workflows followed in the software industry.
- To perform user research, requirement analysis, and competitor analysis.
- To create user personas and user flows.
- To design wireframes and high-fidelity mobile interfaces using Figma.
- To create a reusable design system with consistent typography, colour, spacing, icons, and components.
- To develop an interactive prototype for usability evaluation.
- To improve problem-solving, creativity, communication, teamwork, and documentation skills.

## 1.2 SCOPE OF THE PROJECT

The scope of the EcoGuard – Pollution Control System includes designing the complete experience of monitoring and responding to pollution. The application enables users to register or log in (including guest access), view the live Air Quality Index (AQI) of their location, see individual pollutant levels (PM2.5, PM10, CO, $NO_2$, $SO_2$, $O_3$), find nearby monitoring stations on a map, receive alerts, read health advice, and report pollution incidents with a photo and location. For authorised operators, the application provides a control panel to view sensor status and trigger control actions such as activating air purifiers, water sprinklers, or emission alerts when limits are crossed.

The project focuses primarily on UI/UX design rather than backend or hardware development. It covers the design process from research and requirements to wireframes, visual design, prototyping, and testing. The final interface is intended to provide a clear and consistent experience across common mobile screen sizes.

## 1.3 PROJECT OBJECTIVES

- Present air-quality information in a simple, colour-coded, and understandable manner.
- Provide a clear and predictable navigation structure.
- Give timely alerts together with clear health advice.
- Make incident reporting quick with a minimum number of steps.
- Give operators a clear view of sensors and control actions.
- Maintain visual consistency through reusable components.
- Create a developer-friendly prototype that can be implemented as a mobile application.

## 1.4 BENEFITS OF THE INTERNSHIP

- Practical exposure to industry-oriented UI/UX workflows.
- Hands-on experience with Figma, prototyping, and reusable components.
- Better understanding of user-centered design and data-visualisation design.
- Experience in documenting a complete case study and project report.
- Improved communication, presentation, teamwork, and problem-solving skills.

## 1.5 DESIGN APPROACH

The EcoGuard project follows a user-centered approach in which design decisions are connected to user needs and observed pain points. The process begins with understanding users and existing air-quality applications, continues through definition and ideation, and then moves into wireframing, visual design, prototyping, and testing.

| Stage | Purpose |
|---|---|
| Empathize | Understand user behaviour, expectations, and pain points. |
| Define | Convert research findings into clear design problems. |
| Ideate | Explore alternative layouts, data displays, and interactions. |
| Wireframe | Establish structure and information hierarchy before visual styling. |
| Visual Design | Apply colours, typography, icons, components, and spacing. |
| Prototype | Connect screens and simulate the user journey. |
| Test | Evaluate usability, navigation, readability, and consistency. |

## 1.6 INTERNSHIP OUTCOME

The internship strengthened my practical understanding of UI/UX design and gave me experience in converting user requirements into a structured digital interface. The EcoGuard project improved my confidence in using Figma and in presenting design decisions through research, wireframes, design systems, prototypes, and testing results.

### 1.7 TECHNOLOGIES AND TOOLS

| Technology / Tool | Purpose |
|---|---|
| Figma | UI design, wireframing, components, design system, and interactive prototype. |
| Google Fonts | Typography selection and consistency. |
| Icon libraries | Consistent iconography for air quality, map, alerts, camera, and settings. |
| Image assets | Brand mark, map tiles, and illustrations. |

Figma was the primary design environment. It supported reusable components, Auto Layout, prototyping with connected flows, collaboration, and developer handoff.

## 2.COMPANY PROFILE

### 2.1 ABOUT THE COMPANY

Media Wave Technologies is a software company located in Tiruchirappalli, Tamil Nadu. The company specializes in Web Development, Mobile Application Development, UI/UX Design, Digital Marketing, and Software Solutions. It provides innovative and user-friendly digital solutions for businesses and educational institutions. The company follows industry-standard software development practices and encourages interns to work on real-time projects.

### 2.2 VISION

To become a leading software development company by delivering innovative, reliable, and high-quality digital solutions that enhance customer satisfaction and business growth.

### 2.3 MISSION

- To develop high-quality software applications.
- To provide user-friendly UI/UX design solutions.
- To adopt modern technologies in software development.
- To encourage innovation and continuous learning.
- To deliver excellent customer service.

### 2.4 SERVICES OFFERED

Media Wave Technologies provides the following services:

- Web Application Development
- Mobile Application Development
- UI/UX Design
- Website Design and Development
- Digital Marketing
- Graphic Design
- Software Testing
- Software Maintenance and Support

### 2.5 ORGANIZATION STRUCTURE

The organization follows a structured hierarchy to ensure smooth project execution and effective teamwork.

- Managing Director
- Project Manager
- Team Lead
- UI/UX Designers
- Frontend Developers
- Backend Developers
- Quality Assurance (QA) Engineers
- Interns

The internship was carried out under the guidance of Sivanika S (Software Developer, Industrial Expert and Joint Supervisor) and other experienced professionals, which helped in understanding industry practices, teamwork, project planning, and professional communication.

## 3.DAILY WORK REPORT

The internship lasted for 27 working days from 15 June 2026 to 15 July 2026, during which I learned requirement analysis, user research, wireframing, prototyping, design systems, usability testing, and professional project documentation. The day-wise activities performed during the internship are presented below.

**Day 1:** Company orientation and introduction to the internship project, EcoGuard – Pollution Control System.

**Day 2:** Studied project requirements and understood the objectives of the application.

**Day 3:** Studied air-quality basics: the AQI scale, major pollutants, and health categories.

**Day 4:** Conducted competitor analysis by studying popular air-quality and pollution-monitoring applications.

**Day 5:** Performed user research through interviews and surveys to identify user needs and expectations.

**Day 6:** Created user personas (citizen and control-room operator) based on the research findings.

**Day 7:** Prepared the user flow and information architecture of the application.

**Day 8:** Designed the brand identity: logo, tagline “Breathe. Monitor. Act.” and colour palette.

**Day 9:** Created low-fidelity wireframes for the Login, Sign Up, and Home Dashboard screens.

**Day 10:** Created wireframes for the Pollutant Details, Monitoring Map, and Alerts screens.

**Day 11:** Created wireframes for the Report Pollution and Control Panel screens.

**Day 12:** Reviewed wireframes with the mentor and implemented suggested improvements.

**Day 13:** Created the design system including colours, typography, icons, buttons, cards, and AQI components.

**Day 14:** Designed the Splash, Login, and Sign Up screens using Figma.

**Day 15:** Designed the Home Dashboard with AQI gauge and pollutant cards.

**Day 16:** Designed the Pollutant Details screen with trend graph.

**Day 17:** Designed the Monitoring Map screen with station markers.

**Day 18:** Designed the Alerts screen with colour-coded notification cards.

**Day 19:** Designed the Report Pollution screen with photo, location, and category options.

**Day 20:** Designed the Control Panel screen with sensor status and control actions.

**Day 21:** Designed the Health Advisory screen and the bottom navigation bar.

**Day 22:** Reviewed the complete UI design and ensured consistency across all screens.

**Day 23:** Developed the interactive prototype in Figma by connecting all screens.

**Day 24:** Conducted usability testing for all screens.

**Day 25:** Improved the UI based on testing feedback.

**Day 26:** Verified the layouts for different mobile screen sizes and prepared project documentation.

**Day 27:** Presented the prototype, submitted the final project, and received feedback from the internship mentor.

## 4.BLOCK DIAGRAM

The following block diagram represents the user flow of the EcoGuard application, that is, the steps followed by users while interacting with the application. It helps in creating a smooth navigation experience.

**OPEN APPLICATION (SPLASH SCREEN)**

↓

**LOGIN / SIGN UP / CONTINUE AS GUEST**

↓

**HOME DASHBOARD (LIVE AQI)**

↓

**VIEW POLLUTANT DETAILS / MAP / HEALTH ADVICE**

↓

**RECEIVE ALERT WHEN A LIMIT IS CROSSED**

↓

**REPORT POLLUTION (PHOTO → LOCATION → CATEGORY)**

↓

**OPERATOR: OPEN CONTROL PANEL → CHOOSE CONTROL ACTION**

↓

**TRACK COMPLAINT / ACTION STATUS**

↓

**STATUS RESOLVED (FEEDBACK)**

**Benefits of User Flow**

- Improves navigation.
- Reduces user effort.
- Provides better user experience.
- Helps developers understand application structure.

# 5.ECOGUARD – POLLUTION CONTROL SYSTEM

## 5.1 EXISTING SYSTEM

On many existing platforms, air-quality data is shown as raw numbers with very little explanation, and reporting or controlling pollution requires separate tools. Users often cannot tell what the numbers mean for their health or what action to take.

- Air-quality data is technical and hard to interpret.
- Colour coding and AQI categories are inconsistent between applications.
- Alerts are generic and do not tell users what to do.
- Reporting a pollution incident needs many steps or a separate website.
- Operator dashboards are crowded and difficult to use on mobile.
- Users are forced to create an account before they can view any data.

## 5.2 PROPOSED SYSTEM

EcoGuard aims to provide a centralized experience for monitoring, alerting, reporting, and controlling pollution by dividing the journey into clear and easily understandable stages.

- Large AQI gauge with colour and a one-line meaning.
- Guest access so that anyone can view air quality without signing up.
- Pollutant cards with simple units, safe limits, and trend graphs.
- Map of nearby monitoring stations with colour-coded AQI markers.
- Actionable alerts and health advice for every AQI level.
- Three-step incident reporting (photo, location, category).
- Operator control panel with sensor status and confirm-before-action controls.
- Status tracking for every complaint and control action.

## 5.3 PROPOSED SYSTEM – DESIGN CHARACTERISTICS

| Area | Design Approach |
|---|---|
| Branding | Leaf-and-wind logo, green–teal header with curved edge, calm and trustworthy tone. |
| Navigation | Bottom navigation: Home, Map, Alerts, Profile; back arrow on inner screens. |
| Data Display | AQI gauge, colour-coded pollutant cards, and simple trend graphs. |
| Alerts | Colour-coded cards with severity, time, and recommended action. |
| Reporting | Camera, auto-detected location, category chips, and submit button. |
| Control | Sensor list with status, threshold bars, and confirmation dialogs. |
| Consistency | Reusable components and common design tokens. |

### 5.4 CAPABILITIES

- User registration and login with Google, Apple, and email options.
- Guest access ("Continue as guest").
- Live AQI and pollutant levels (PM2.5, PM10, CO, $NO_2$, $SO_2$, $O_3$) for the current location.
- Monitoring station map with AQI markers.
- Alerts with health advice for different groups of people.
- Pollution incident reporting with photo and location.
- Operator control panel for sensor status and control actions.
- Trend and history view for the last 24 hours and 7 days.

## 6.PROPOSED METHOD

### 6.1 USER RESEARCH

User research is the process of understanding user behaviour, needs, expectations, and problems before designing an application. It helps designers create solutions that are useful and user-friendly. For the EcoGuard – Pollution Control System, user research was conducted to understand how people check air quality and what difficulties they face while doing so.

**Research Methods**

- User interviews
- Competitor analysis
- Online surveys
- Observation of existing applications

**Research Findings**

- Users want to understand air quality at a glance without technical knowledge.
- Users want to know what to do when air quality is poor.
- Users prefer location-based data and nearby station information.
- Users want a fast way to report smoke, dust, burning, or industrial emissions.
- Operators need a clear and uncluttered view of sensors and thresholds.
- Users dislike being forced to sign up before viewing data.

### 6.2 USER PERSONA

A user persona is a fictional representation of a target user created based on research data. It helps designers understand user goals, needs, and challenges.

**Persona 1 – Citizen**

Name: Divya Ramesh | Age: 22 Years | Occupation: College Student

**Goals:** Check air quality before going out; get health advice; report pollution. **Needs:** Simple data, clear alerts, quick reporting. **Pain Points:** Confusing numbers, generic alerts, long reporting forms.

**Persona 2 – Operator**

Name: Suresh Kumar | Age: 38 Years | Occupation: Control-Room Operator

**Goals:** Monitor sensors; respond quickly when limits are crossed. **Needs:** Clear status indicators, one-tap actions, action history. **Pain Points:** Crowded dashboards, delayed alerts.

## 6.3 WIREFRAMES

Wireframes are basic layouts that represent the structure and arrangement of elements in an application before creating the final design. Wireframing helps in planning screen layouts, navigation, and user interactions.

**Wireframe Screens Created**

- Splash Screen
- Login and Sign Up Screens
- Home Dashboard
- Pollutant Details Screen
- Monitoring Map Screen
- Alerts Screen
- Report Pollution Screen
- Control Panel Screen
- Health Advisory Screen

**Advantages**

- Easy design modifications.
- Better planning of user interface.
- Identifies usability issues early.
- Improves communication between designers and developers.

## 6.4 DESIGN SYSTEM

A design system is a collection of reusable design elements that maintain consistency throughout the application. The EcoGuard design system includes:

**Colour Palette**

- Primary Colour – Deep Green / Teal
- Background Colour – Soft White
- Text Colour – Dark Grey
- Secondary Colour – Light Grey
- AQI Scale – Good (Green), Satisfactory (Yellow-Green), Moderate (Yellow), Poor (Orange), Very Poor (Red), Severe (Maroon)

**Typography**

- Headings and Buttons – Bold sans-serif
- Body Text and Labels – Clean sans-serif, Regular and Medium
- AQI Values – Large numerals for quick reading

**UI Components**

- Pill buttons and login / sign up tab switch
- Input fields with show-password icon
- AQI gauge and pollutant cards
- Station markers and station card
- Alert cards and category chips
- Sensor status rows and confirmation dialogs
- Bottom Navigation Bar

**Benefits**

- Maintains consistency.
- Saves design time.
- Improves user experience.
- Provides reusable components.

## 6.5 HIGH-FIDELITY UI DESIGN

High-fidelity UI design represents the final visual appearance of the application with complete colours, images, typography, icons, and interactive elements. The high-fidelity screens of EcoGuard were designed using Figma by applying modern UI principles.

**Design Principles Followed**

- Simplicity
- Consistency
- Accessibility (sufficient contrast, and text labels along with colour)
- Visual hierarchy
- Responsive design

## 6.6 PROTOTYPE

A prototype is an interactive model of the application that demonstrates how users navigate between different screens. The EcoGuard prototype was created using Figma by connecting all designed screens and adding interactions, as shown in Figure 9.11.

**Prototype Features**

- Screen navigation
- Button interactions
- Alert and reporting flow demonstration
- User journey simulation

**Advantages**

- Helps evaluate user experience.
- Identifies design improvements.
- Provides realistic application preview.
- Supports developer understanding.

# 7.ALGORITHM

## 7.1 USER LOGIN AND REGISTRATION

- User opens the app and the Splash screen is shown.
- User lands on the Login screen.
- User enters email and password, or selects Google or Apple sign-in.
- System checks whether the required fields are completed.
- If credentials are valid, the user is taken to the Home Dashboard.
- If the user is new, the Sign Up tab opens registration (name, email, phone, new password, confirm password).
- Registration data is validated, including password match, before account creation.
- Alternatively, the user selects “Continue as guest” and goes directly to the Home Dashboard.

## 7.2 AIR QUALITY MONITORING

- The Home Dashboard detects the user’s location and shows the live AQI with its colour and category.
- Pollutant cards show PM2.5, PM10, CO, $NO_2$, $SO_2$, and $O_3$ values.
- User taps a pollutant to see its trend graph and safe limit.
- User opens the Map to view nearby monitoring stations and their AQI.

## 7.3 ALERTS AND HEALTH ADVICE

- When a pollutant crosses its threshold, an alert is generated with severity and time.
- The Alerts screen lists all alerts, newest first.
- User taps an alert to view the Health Advisory for that AQI level.

## 7.4 REPORT POLLUTION

- User taps “Report Pollution”.
- User captures or uploads a photo; the location is auto-filled and can be edited.
- User selects a category (smoke, dust, burning, industrial emission, noise).
- User submits the report; a complaint ID and status (Received, Under Review, Action Taken, Resolved) are shown.

## 7.5 CONTROL ACTIONS (OPERATOR)

- Operator logs in and opens the Control Panel.
- The sensor list shows the status and threshold bar of each sensor.
- For a sensor above its limit, the operator selects an action (activate air purifier, start sprinklers, send emission alert).
- A confirmation dialog is shown; on confirmation, the action status is displayed and logged.

## 7.6 GENERAL DESIGN ALGORITHM

Requirement → Research → Problem Definition → Ideation → Information Architecture → Wireframe → Design System → High-Fidelity UI → Prototype → Testing → Feedback → Refinement → Final Handoff

# 8.DESIGN IMPLEMENTATION

## 8.1 SPLASH SCREEN

The Splash Screen introduces the brand with the EcoGuard logo – a leaf with wind lines inside a circle – on a deep green background with the tagline “Breathe. Monitor. Act.”

## 8.2 LOGIN PAGE

The Login Page allows users to access the application using email and password. It contains a login / sign up tab switch, a show-password icon, a Forgot Password link, Google and Apple sign-in options, and a “Continue as guest” button.

## 8.3 SIGN UP PAGE

The Sign Up Page collects name, email, phone number, new password, and confirm password, with show-password icons and a prominent Sign Up button.

## 8.4 HOME DASHBOARD

The Home Dashboard shows the current location, a large AQI gauge with category and advice, a horizontal row of pollutant cards, a 24-hour trend, and a quick “Report Pollution” button. The bottom navigation bar gives access to Home, Map, Alerts, and Profile.

## 8.5 POLLUTANT DETAILS PAGE

This page shows the selected pollutant’s current value, safe limit, health effect, and a 24-hour / 7-day trend graph.

## 8.6 MONITORING MAP PAGE

The Map Page shows nearby monitoring stations as colour-coded AQI markers, with a station card showing name, AQI, last updated time, and distance.

## 8.7 ALERTS PAGE

The Alerts Page lists alerts as colour-coded cards showing severity, location, time, and recommended action.

## 8.8 REPORT POLLUTION PAGE

The Report Pollution Page provides photo upload, auto-detected location, category chips, an optional description, and a Submit button, followed by a confirmation with complaint ID and status timeline.

## 8.9 CONTROL PANEL PAGE

For operators, the Control Panel shows a sensor list with status, threshold bars, control action buttons, a confirmation dialog, and action history.

## 8.10 HEALTH ADVISORY PAGE

The Health Advisory Page shows advice for the current AQI level for the general public, children, the elderly, and people with respiratory conditions.

## 9.SAMPLE OUTPUT

The following sample outputs are taken from the EcoGuard UI/UX project and demonstrate the final high-fidelity interface screens developed during the internship. The AQI values, station names, and readings shown in the screens are sample data used only for demonstration.




*Figure 9.1 – Splash Screen*

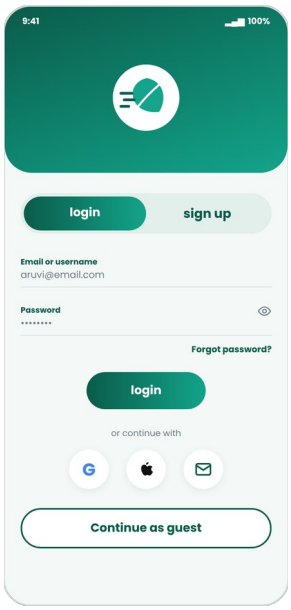



*Figure 9.2 – Login Page*

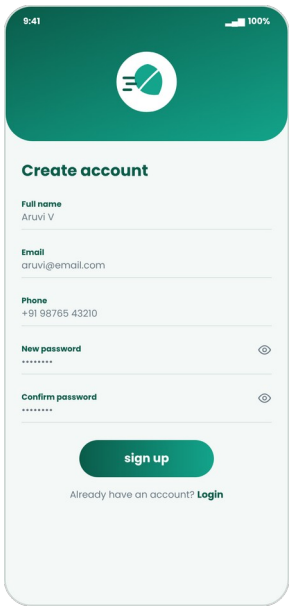



*Figure 9.3 – Sign Up Page*

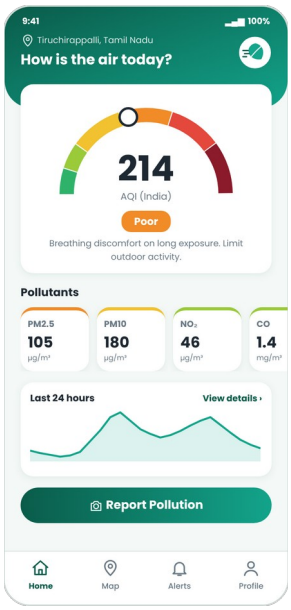



*Figure 9.4 – Home Dashboard*

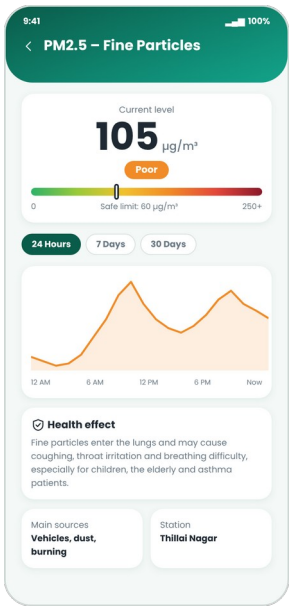



*Figure 9.5 – Pollutant Details Page*

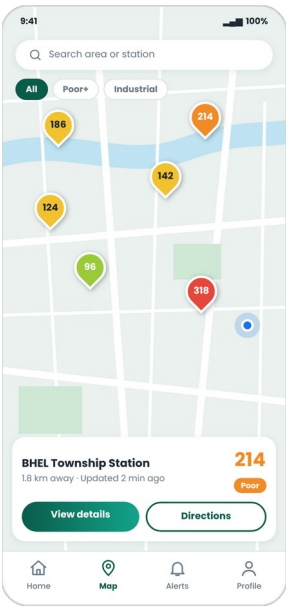



*Figure 9.6 – Monitoring Map Page*

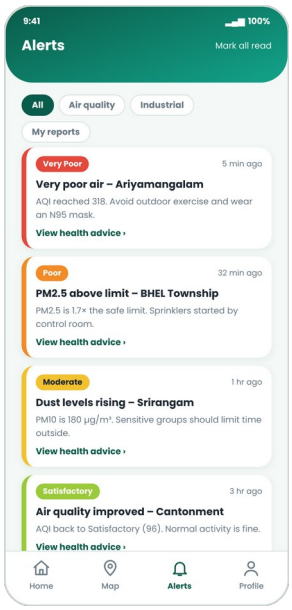



*Figure 9.7 – Alerts Page*

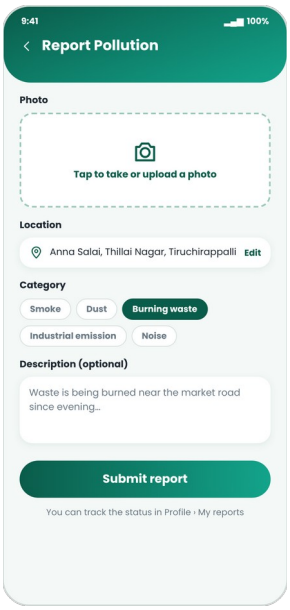



*Figure 9.8 – Report Pollution Page*

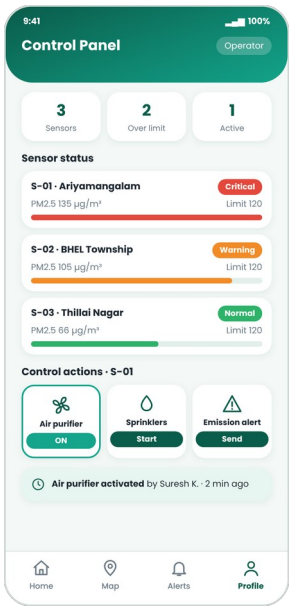



*Figure 9.9 – Control Panel Page*

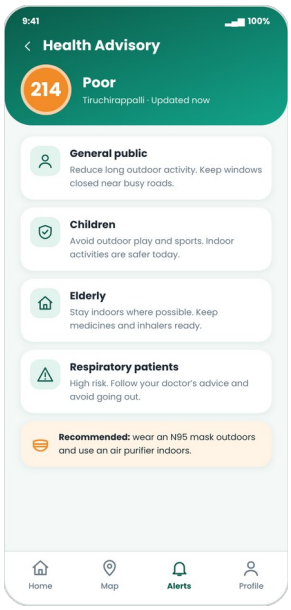



*Figure 9.10 – Health Advisory Page*

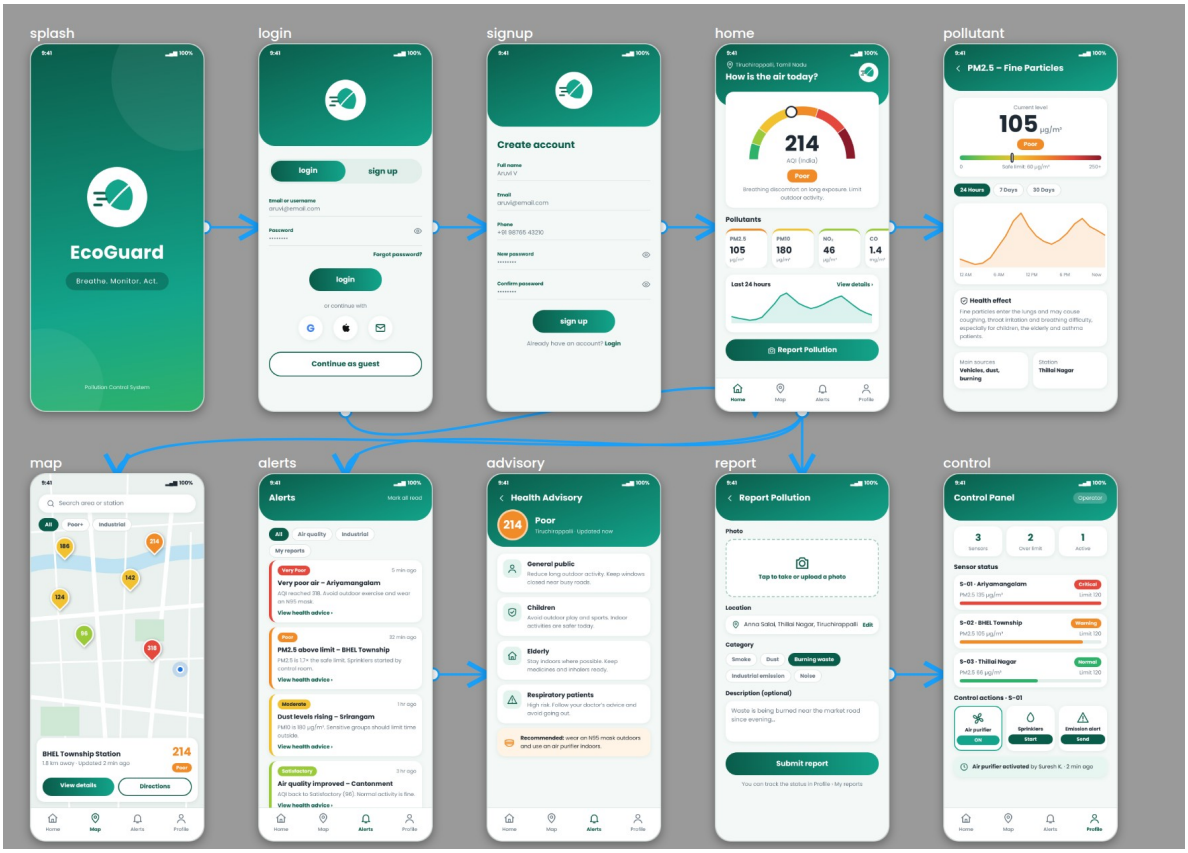



*Figure 9.11 – Complete Prototype Flow in Figma*

# 10.ANALYSIS

## 10.1 TESTING OVERVIEW

Testing is an important phase in the software development process used to verify the quality, usability, and performance of the application. The EcoGuard interface was tested to ensure smooth navigation, readable data, consistency, and user-friendly interaction.

## 10.2 FUNCTIONAL TESTING

Functional testing was performed to verify whether all prototype features work according to the requirements.

**Tested Features**

- Login, sign up, and guest access flow
- Navigation between pages and bottom navigation
- AQI and pollutant viewing
- Monitoring map interaction
- Alerts and health advisory access
- Incident reporting flow
- Control action with confirmation dialog

## 10.3 UI TESTING

UI testing was conducted to check the visual appearance and consistency of the application.

**UI Testing Includes**

- Colour consistency and AQI colour accuracy
- Font readability
- Button alignment and tap-target size
- Icon and image placement
- Layout structure
- Responsive design on different mobile screen sizes

## 10.4 USABILITY TESTING

Usability testing was performed to evaluate the ease of use of the EcoGuard application.

**Testing Objectives**

- Check user navigation flow.
- Identify design issues.
- Improve user interaction.
- Ensure simple and effective interface design.

## 10.5 TEST RESULTS

| Test Area | Result |
|---|---|
| Splash, Login and Sign Up | Passed |
| Home Dashboard Navigation | Passed |
| Pollutant Details and Trends | Passed |
| Monitoring Map | Passed |
| Alerts and Health Advisory | Passed |
| Report Pollution | Passed |
| Control Panel | Passed |
| Responsive Design | Passed |

# 11.CONCLUSION

## 11.1 LEARNING OUTCOMES

The internship at Media Wave Technologies, Tiruchirappalli provided practical experience in UI/UX design and real-time project development.

**Skills Gained**

- UI/UX Design principles and user research
- Wireframing and prototyping
- Design system creation and data-visualisation design
- Figma tool usage
- Usability testing
- Professional communication and team collaboration

## 11.2 CONCLUSION

The internship provided valuable industrial exposure and helped me understand the complete UI/UX design process. The EcoGuard – Pollution Control System was successfully designed by applying user-centered design principles.

Through this internship, I gained practical knowledge in creating modern, responsive, and user-friendly mobile interfaces. The experience improved my technical skills, creativity, and confidence in handling real-world software projects.

The EcoGuard design brings live air-quality data, health guidance, incident reporting, and operator controls into one calm and consistent mobile experience. By applying Design Thinking, user research, a reusable design system, and iterative usability testing, the project delivered an interface that is clear for citizens and efficient for control-room operators. The prototype can now serve as a reliable reference for frontend and backend development.

**Key Outcomes of the Project**

- A complete set of ten high-fidelity screens covering the citizen and operator journeys.
- A reusable design system with colour, typography, icon, and component standards.
- An interactive Figma prototype that demonstrates the end-to-end user journey.
- Functional, UI, and usability test results confirming clear navigation and visual consistency.
- Complete project documentation prepared for developer handoff.

## 12.FUTURE WORK

The future enhancements of the EcoGuard application include:

- AI-based pollution forecasting.
- Voice-based queries such as “What is the air quality today?”.
- Personalised health alerts based on user conditions.
- Multi-language support, including Tamil.
- Community reports map and complaint upvoting.
- Dark mode and accessibility improvements.
- Frontend development and backend / IoT sensor integration of the design.
- Offline access to the most recent air-quality readings.
- Exportable history and reports for operators and authorities.

These enhancements will extend EcoGuard from a design prototype into a complete, data-driven pollution monitoring and response platform that serves both citizens and authorities.

## 13.REFERENCES

The following references were used for understanding UI/UX design principles, software development methodologies, and design tools.


1. Figma. (2026). Figma Documentation. Available at: https://help.figma.com/
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Media Wave
TECHNOLOGIES

## INTERNSHIP OFFER LETTER

**03rd Jun 2026**

Dear **Ms. ARUVI V**

We are pleased to offer you an internship with **MEDIA WAVE TECHNOLOGIES.** This is an internship. Our goal is for you to learn more about **UI/UX DESIGN.**

As we discussed, your internship is expected to last Commence from **June 15, 2026, to July 15, 2026**. However, at the sole discretion of the Company, the duration of the internship may be extended or shortened with or without advance notice.

As an intern, you will not be a Company Employee. Therefore, you will not receive a salary, wages, or other compensation. In addition, you will not be eligible for any benefits that the Company offers its employees.

During your internship, you may have access to confidential, proprietary, and/or trade secret information belonging to the Company. You agree that you will keep all of this information strictly confidential and refrain from using it for your own purposes or from disclosing it to anyone outside the Company. In addition, you agree that, upon conclusion of the internship, you will immediately return to the Company all of its property, equipment, and documents.

Your responsibilities may include assisting in UI/UX design, wireframing, prototyping, user research, interface design, usability testing, and related creative activities.

Sincerely,

**Chandru S**
**(HR Department)**

Media Wave Technologies
Trichy

WD-54, Anandha bhavan complex,
Second floor, 17/52, Puthur High Rd,
Tiruchirappalli, Tamil Nadu 620017

+91 6369153235

mediawavetech.com
support@mediawavetech.com

Media Wave
TECHNOLOGIES

## Internship Completion Letter

**July 15, 2026,**

**Strictly Confidential**

**To Whom It May Concern**

This letter certifies that **Ms. ARUVI V** has interned in the role of **UI/UX Design** with **Media Wave Technologies** during the period **15 June-2026 to 15 July-2026.**

While with Media Wave Technologies, she has remained dedicated and loyal to her work and responsibilities with our company. She has done an exemplary job while in this **UI/UX Design role**. She has always maintained a professional and courteous attitude and appearance while with our company.

Her decision to end her internship with our company is solely her own decision, and we wish her all the best in her future career opportunities.

If you have any questions, please clarify from the undersigned.

Sincerely,

**Chandru S**
**(HR Department)**

WD-54, Anandha bhavan complex,
Second floor, 17/52, Puthur High Rd,
Tiruchirappalli, Tamil Nadu 620017

+91 6369153235

mediawavetech.com
support@mediawavetech.com

Media Wave TECHNOLOGIES

## ATTENDANCE CERTIFICATE

**To,**

The Head of the Department, Assistant professor,
University College of Engineering (BIT Campus),
Anna University,

Tiruchirappalli-620024.

**Dear Sir/MAM,**

This is to certify that **Ms. ARUVI V** maintained excellent attendance during the period from **15th June 2026 to 15th July 2026** while undergoing on **UI/UX DESIGN Internship Training** conducted at **MEDIA WAVE TECHNOLOGIES.**

**The listed below status of attendance:**

- **Total Working Days:** 27 Days
- **Number of Days Present:** 27 Days
- **Attendance Percentage:** 100%

During this period, she has shown sincere interest, punctuality, and good conduct. We appreciate his commitment and dedication.

With best regards,

Chandru S
HR Department

Media Wave Technologies Trichy

WD-54, Anandha bhavan complex,
Second floor, 17/52, Puthur High Rd,
Tiruchirappalli, Tamil Nadu 620017

+91 6369153235

mediawavetech.com
support@mediawavetech.com

## ANNEXURE

**Internship (June – July 2026)**

**1.Name of the Industry with address:**
Media wave Technologies -WD 54,Ananda Bhavan Complex,
Second floor,17/52,Puthur High Road, Tiruchirappalli -620017

**2.Name of the Industrial Expert (Joint Supervisor):** Sivanika S

**3.Qualification of the Industrial Expert (Joint Supervisor):** Bachelor Degree

**4.Designation of the Industrial Expert (Joint Supervisor):** Software Developer

**5.Title of the Project Work:**

Pollution Control System -Mobile Application

**6.Duration of the Internship:** 15/06/2026 – 15/07/2026

**7.Details of the student:**

| Sl. No. | Name of the student | Registration No. | Department |
|---|---|---|---|
| 01 | ARUVI V | 810023104027 | Computer Science and Engineering |

**Terms and conditions**

- The student is permitted to undergo a semester long internship in your organization
- The student shall be instructed to meet the Institute supervisor periodically and to attend the review committee meetings for evaluating the progress.
- The final report of the internship duly signed by the industrial expert/supervisor will be submitted to college.
- Attendance Certificate in the prescribed format signed by the competent authority of the industry will be produced
- Internship Certificate in the prescribed format signed by the competent authority of the industry will be produced

Media Wave Technologies Trichy

Seal of the Organization

Signature

Name and Designation of the Officer with seal